Observation

RGB-D Images

Tracking

Past SE(3) Trajectory

Anchor Image

Trellis-1

Scale Estimation

Scaled Mesh

VLM

Material: **wood**
Density: $0.4g/cm^3$

Neural SE(3) Forecaster

History Encoder (GRU)

Dynamics context

Velocity Estimator (GRU)

Initial velocities

$\tilde{v}_t^W,\ \omega_t^B$

Assemble initial state

$z_t = [\tilde{p}_t^W, \tilde{v}_t^W, \tilde{q}_t, \omega_t^B]$

Neural ODE Dynamics

Acceleration Field

$A_\theta(z_t, c_t, \psi) = [a_t^W, \alpha_t^B]$

$\dot{z} = F_\theta(z, c_t, \psi)$

Newton-Euler structured state space transition

ODE Solver odeint

Future SE(3) Trajectory

Inertia Prior

$\psi = I^B$

Interception Selector

Selection objective

$$k^* = \arg\min_{k \in \mathcal{K}_{\text{feas}}} J(k), \quad g^* = g_{k^*}$$

$$J(k) = \begin{cases} T_{\text{reach}}(k), & \text{fast,} \\ |s_k - \alpha|, & \text{slow.} \end{cases}$$

Target EEF g^*

Track

Forecast

Act

Success!

Training in Simulation w. Primitives

GENESIS